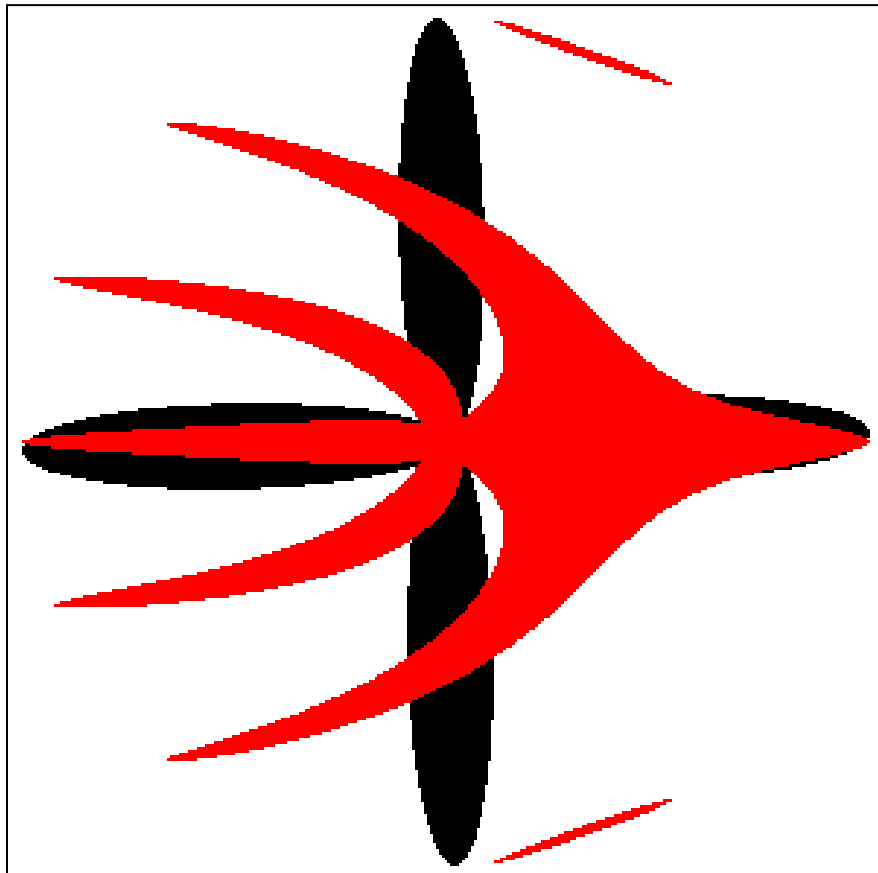


chapter 8 - camera shutter effect



This script uses the NumPy, Tkinter, and Pillow libraries to create a graphical animation demonstrating the rolling shutter effect on a rotating four-blade propeller. It mathematically models the spinning blades using a 2D complex number grid and simulates a digital camera sensor by scanning downward, capturing only a single horizontal row of pixels during each frame. As the loop progresses, it combines the real-time rotating propeller with the progressively captured rows to render a composite image, visually explaining why fast-moving objects appear bent or warped in digital photography.